

Treatment of gravitational forces in an open-chain manipulator

On p. 170 MLS defines the i -th row of the column vector N as

$$-N_i(\theta, \dot{\theta}) = -\frac{\partial V}{\partial \theta_i} - \beta \dot{\theta}_i$$

and then, on p. 171, they indicate the use of N in the equations of motion of an open-chain manipulator:

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = \tau \quad (4.24)$$

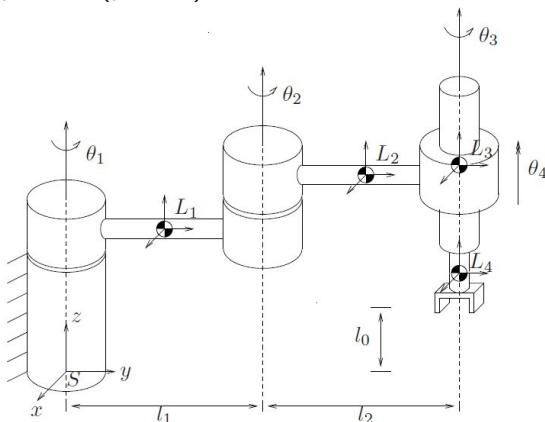
Note that N_i in general depends on all n joint angles and angular velocities of the manipulator (not only the i -th).

$$V(\theta) = \sum_{i=1}^n V_i(\theta) = \sum_{i=1}^n m_i g h_i(\theta)$$

on p. 169 rightly indicate that h_i is in general a function of all n joint angles.

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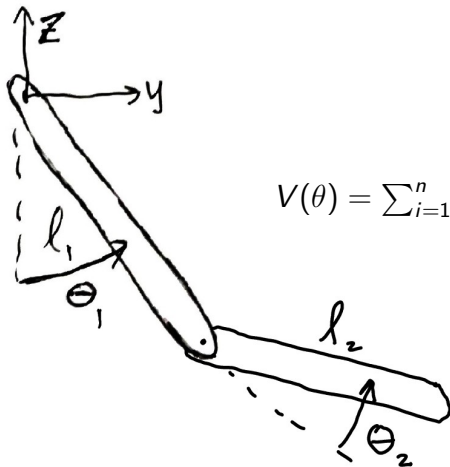
MLS Example 4.4 (p. 177)



is not a good example, in this regard.

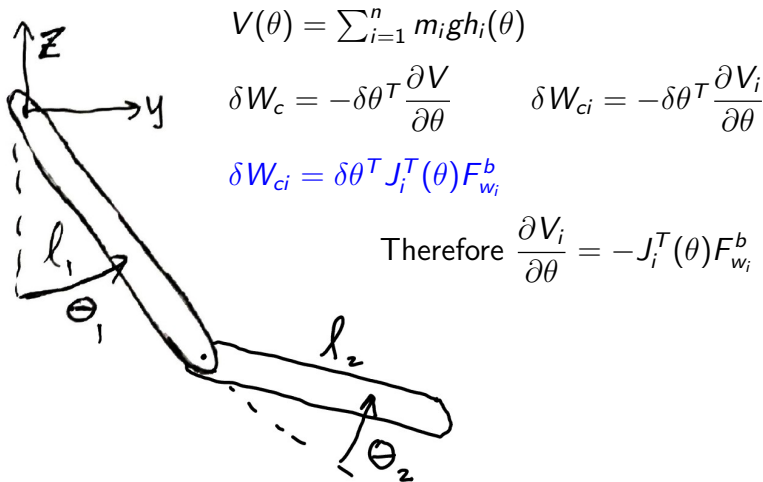
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Better example:



$$V(\theta) = \sum_{i=1}^n m_i g h_i(\theta)$$

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Treatment of gravitational forces in an open-chain manipulator

$$\frac{\partial V_i}{\partial \theta} = -J_i(\theta)^T F_{w_i}^b$$

$$J_i(\theta) = [\xi_1^\dagger \quad \cdots \quad \xi_i^\dagger \quad 0 \quad \cdots \quad 0]$$

$$\xi_j^\dagger = \begin{cases} \text{Ad}_{\left(e^{\hat{\xi}_{j+1}\theta_{j+1}} \cdots e^{\hat{\xi}_i\theta_i} g_{sl_i}(0)\right)}^{-1} \xi_j & j < i \\ \text{Ad}_{\left(g_{sl_i}(0)\right)}^{-1} \xi_j & j = i \end{cases}$$

as on MLS pp. 168 and 176

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$$\frac{\partial V_i}{\partial \theta} = -J_i(\theta)^T F_{w_i}^b$$

Typically

$$F_{w_i}^s = [0 \quad 0 \quad -m_i g \quad 0 \quad 0 \quad 0]^T$$

$$F_{w_i}^b = \text{Ad}_{g_{sl_i}}^T F_{w_i}^s \approx (2.67) \text{ (p. 63)}$$

Typically, 1st 3 elements of 3rd row of g_{sl_i} multiplied by $-m_i g$ form 1st 3 elements of $F_{w_i}^b$ (rest 0).

Virtual displacement calculation

Body velocity of frame L_i :

$$V_{sl_i}^b = J_{sl_i}^b(\theta)\dot{\theta} \quad \text{or} \quad V_i^b = J_i(\theta)\dot{\theta}$$

where $J_{sl_i}^b = J_i$ is the body Jacobian corresponding to g_{sl_i}

MLS p. 61

$$\Delta W = V_{ab}^{bT} F_b \Rightarrow \Delta W_i = V_i^{bT} F_w^b = \dot{\theta}^T J_i^T(\theta) F_w^b$$

$$\delta W = \delta\theta^T J_i^T(\theta) F_w^b$$